#5 · 9.6% of marks

Ch 5 — Security Valuation

SM order: return concepts → ERP → required return → discount-rate selection → equity (dividend / earnings / cash-flow models, EV, rights) → preference → bonds → money market → valuers → precautions. TYK theory Qs: **why duration < maturity** (8.8) · **zero coupon bonds** (8.15).

① Return concepts (SM 2)

- → ***Required rate of return*** — *minimum return the investor expects over a period; also called **opportunity cost / cost of capital** (highest expected return forgone elsewhere at similar risk). Often used interchangeably with expected return.*
- → ***Discount rate*** — *rate used to find PV of future cash flows = risk-free rate + risk premium. Each cash-flow stream can carry a different rate (inflation, maturity, default probability → term structure: upward sloping = longer maturity, more inflation / liquidity / default risk).*
- → ***IRR*** — *rate equating PV of future cash flows to market price; average annual return **assuming reinvestment at the IRR**. SM eg: invest 20,000 → 3,000 (yr 1) + 23,000 (yr 2) → IRR **15%**; if the 3,000 is reinvested at only 10% → $(26{,}300/20{,}000)^{1/2} - 1$ = **14.67%**.*

② Equity risk premium vs market risk premium (SM 3.2) ⚠

CAPM: $R_x = R_f + \beta_x(R_m - R_f)$

Market Risk Premium = $R_m - R_f$ · **Equity Risk Premium = $R_x - R_f = \beta_x(R_m - R_f)$**

If x = market (β = 1) → ERP = market premium.

> ⚠ **TRAP**
> Earlier sheet wrongly said "ERP = $R_m - R_f$". Per SM, that is the **market** risk premium; ERP of a stock is **β × MRP**.

③ Required return on equity (SM 4)

Return on share A = R_f + β × Market Risk Premium. CAPM's insight: investors judge risk by the asset's contribution to **systematic** risk (not diversifiable); gives a relatively objective, widely used estimate. SM eg: 5% + 1.5 × 4.5% = **11.75%**. β from covariance (Ch 6): β = $Cov_{im} \div \sigma_m^2$. PYQ May 2024: 30 ÷ 24 = **1.25** → k_e = 12 + 1.25 × 4 = **17%**.

④ Discount rate selection (SM 5)

- → *Cash flows to **equity** → required return on equity; cash flows to **all stakeholders** → cost of capital (WACC)*
- → ***Nominal** CF = not adjusted for inflation; **real** CF = adjusted. Nominal CF ↔ nominal rate; real CF ↔ real rate. Low inflation / short term → nearly same; high inflation → nominal > real.*
- → *Equity valuation uses **only nominal** CF and nominal rate — because corporate tax is stated in nominal terms, so nominal CF reflects taxes accurately. Equity CF → nominal k_e; all-provider CF → **nominal after-tax WACC**.*

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5 Dividend-based models (SM 6.1)

Assumptions: (a) dividend paid annually; (b) first dividend at end of year 1; (c) shares sold at year-end at **ex-dividend** price.

Model	Formula (SM)
One-year holding	$P_0 = (D_1 + P_1) \div (1 + K_e)$ · SM eg: (6 + 36) ÷ 1.20 = **₹35**
Zero growth	$P_0 = D \div K_e$
Constant growth	$P_0 = D_1 \div (K_e - g) = D_0(1 + g) \div (K_e - g)$ — based on Gordon growth model of K_e
Two-stage	$\Sigma\ D_0(1 + g_1)^t \div (1 + K_e)^t$ for n yrs + $P_n \div (1 + K_e)^n$; **$P_n = D_0(1 + g_1)^n(1 + g_2) \div (K_e - g_2)$**
Three-stage	Extraordinary growth → **transition** (advantage depleted by competition) → stable; PV of dividends of 2 periods + PV of price at start of stable period
H-model	$P_0 = D_0(1 + g_n) \div (r - g_n) + D_0 \times H \times (g_c - g_n) \div (r - g_n)$; growth declines **linearly over 2H**; H = half the transition period; = value at normal growth + premium for abnormal growth

SM: one growth rate must be assumed **for infinity**, only then can value be found. These variants also apply to the FCFE model.

Worked (app-computed, re-checked in code)	Result
Constant growth: D_0 3, g 8%, K_e 17%	3.24 ÷ 0.09 = **₹36.00**
MTP Jan 2026: R_f 8, R_m 12, β 1.5 → K_e 14%; D_0 3, g 10%	3.30 ÷ 0.04 = **₹82.50**
H-model: D_0 2, g_c 16%, g_n 6%, transition 10 yrs (H 5), r 12%	(2.12 + 2 × 5 × 0.10) ÷ 0.06 = **₹52.00**
Two-stage: D_0 2, 20% for 3 yrs then 6%, K_e 12%	PV divs 6.90 + PV P_3 43.46 = **₹50.36**
PYQ May 2024: D_0 3, 30% for 5 yrs then 15%, K_e 17%	20.81 + 292.09 = **₹312.90**

> ⚠ **TRAP**
> P_n uses **D_{n+1}** (grow the last dividend once more at g_2) and is discounted by the **year-n** factor. Buy if intrinsic > market; hold if roughly equal.

6 Earnings-based models (SM 6.2)

Investors may forgo cash dividend for higher earnings on retained profits → value on earnings.

(a) Gordon's model — assumes (i) return on retained earnings constant, (ii) retention ratio constant: **$P = EPS_1(1 - b) \div (K_e - br)$** (r = ROE, b = retention)

(b) Walter's approach: **$P = [D + (E - D) \times r/K_e] \div K_e$**

(c) P/E (multiplier) approach: Value = EPS × P/E; P/E estimated from similar company / industry after adjusting for company-specific features

SM TYK 1: BVPS 137.8, ROE 15%, retention 60%, K_e 18% → EPS 20.67, D 8.27, g = 0.6 × 0.15 = 9% → Gordon **₹91.89**; Walter = [8.27 + (12.40) × 0.15/0.18] ÷ 0.18 = **₹103.35**.

REPEAT R11

⑦ Free cash flow models (SM 6.3)

DDM ignores cash needed for long-term capex and short-term working capital → does not reflect true free cash. **FCFF → discount at K_0** (overall firm value); **FCFE → discount at K_e**. **FCFE along with DDM values equity**; FCFF values the firm.

FCFF based on	Formula (SM 6.3.1)
(a) Net income	NI + Interest(1 − t) + Dep ∓ Capex ∓ Δ non-cash NWC
(b) EBIT	EBIT(1 − t) + Dep ∓ Capex ∓ Δ non-cash NWC
(c) EBITDA	EBITDA(1 − t) + **Dep × t** ∓ Capex ∓ Δ non-cash NWC
(d) FCFE	FCFE + Interest(1 − t) + Principal repaid − New debt + Preferred dividend
(e) CFO	CFO + Interest(1 − t) ∓ Capex (US GAAP; don't add interest again if already in CFO)

FCFE (SM 6.3.2) = Net Profit + Dep − ΔNWC − Capex + New Debt − Debt Repayment **+ Net issue of Preference Shares − Preference Dividends**. Can also be used in the multistage growth model.

→ Sep 2025: FCFF with WACC from a **target** structure → equity value per share; then FCFE back-solved

→ May 2026: two-stage FCFE; **implied terminal growth** for a given market price

⑧ DDM vs FCFE (SM 6.4)

DDM	FCFE
Values stock on cash paid as dividend; assumes consistent payout, which can be **less than cash available**. Healthy cash + low payout → **undervaluation**.	Values stock on cash available for distribution **after** capex and incremental WC → **better measure** than DDM.

⑨ Enterprise value (SM 6.5)

EV = Market cap + LT debt + Minority interest − Cash & equivalents (also minus equity investments in affiliates / other companies and LT investments)

Type	Meaning
Total EV	All business activities: mkt cap + interest-bearing debt + minority interest − cash
Operating EV	Total EV − market value of **non-operating** assets (investments, shares in associates)
Core EV	Operating EV − value of **non-core** assets

EV = true economic value; considers debt + equity, **least affected by capital structure** → EV multiples more reliable than equity multiples. **EV/Sales** — firms with negative CF / earnings (cyclical, tech); sales least manipulable. **EV/EBITDA** — capital-intensive firms (heavy D&A), acquisitions; unaffected by depreciation policy & capital structure; inverse = cash return on total investment.

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10 Rights (SM 6.6)

TERP = $(nP_0 + S) \div (n + n_1)$ · **Value of right** = Ex-right price − Subscription price · **Value per shareholding** = (Ex-right − Subscription) ÷ existing shares per right

SM TYK 2: ₹13, 10 lakh shares, raise ₹20 lakh. 1-for-2 → 5 lakh @ ₹4 → TERP **₹10**, right **₹6**. 1-for-4 → 2.5 lakh @ ₹8 → TERP **₹12**, right **₹4**.

11 Preference shares (SM 7)

→ Fixed dividend like debentures. **Non-redeemable** = like perpetual bond: Dividend ÷ required return

→ **Redeemable** = PV of dividends + PV of maturity value at required return. SM eg: FV 10,000, 10%, 3 yrs, 12% → **₹9,519.63**

12 Bond basics, value, theorems (SM 8.1–8.7)

→ **Basics**: par value · coupon rate & frequency (monthly / quarterly / half-yearly / annual) · maturity period · redemption (**bullet** — one-shot at par or premium)

→ **$V = \Sigma\ I \div (1 + k_d)^t + F \div (1 + k_d)^n$** = I(PVIFA) + F(PVIF); k_d = YTM / required return on similar bonds

→ **Semi-annual**: V = I/2 × PVIFA(k_d/2, 2n) + F × PVIF(k_d/2, 2n)

→ **YTM** = discount rate at which PV of the bond's future cash flows = market price

Cause (SM 8.3)	Effect
YTM = coupon rate	sells at **par**
YTM > coupon rate	sells at **discount**
YTM < coupon rate	sells at **premium**
Longer maturity	greater price change for a given change in required return

→ **Price–yield (8.6)**: price varies **inversely** with yield

→ **Price–time (8.7)**: price = par at maturity (no default) → premium bond falls over time, discount bond rises

Worked (app-computed): Face 1,000, coupon 10%, 5 yrs, k_d 12% → **₹927.90**. Exam aid, not SM: approx YTM = [C + (RV − P) ÷ n] ÷ [(RV + P) ÷ 2].

13 Duration (SM 8.8)

Duration = weighted average time within which investor gets back promised principal and promised YTM; measures how quickly a bond repays its true cost; also a measure of **interest-rate risk** (higher duration → higher risk).

Coupon bond's duration is **always less than its maturity** (TYK theory Q1).

Factors: (i) **time to maturity** — shorter → lower duration; (ii) **coupon** — higher → lower; (iii) **YTM** — higher → lower

Macaulay = $\Sigma\ t \times PV(CF_t) \div$ price (years to recover true cost) · **Modified** = Macaulay ÷ (1 + YTM/n), n = compounding periods per year

Same bond (app-computed): D = **4.1355 yrs**; MD = **3.6924** → +1% yield ≈ **−3.69%** price.

13 Immunization (SM 8.9) — contd.

- → **Immunization**: interest-rate risk = **price risk + reinvestment risk**, which move in **opposite** directions. Choose bonds so they offset → portfolio value at end of holding period insensitive to rate changes. **Duration = holding period** → immunized.
- → SM TYK 27: need ₹1 lakh in 2 yrs; Bond X duration 1, Bond Y 3.563 → 2 = x + (1 − x)3.563 → **61% X, 39% Y** (≈ 52 X, 34 Y bonds)

14 Yield curve / term structure (SM 8.10–8.11)

Spot rate = rate for a zero-coupon bond of that maturity (time value of a single future payment). **Yield curve** (term structure) = how YTM relates to term to maturity, for bonds similar in all respects except maturity. **Four patterns** (Saransh): **A. Upward sloping · B. Downward sloping · C. Flat · D. Humped**. Forward rates (rates applicable in future) are derived from spot rates.

Theory	Says (SM)
Expectation	Long-term rates are **unbiased estimators** of future short-term rates → used to forecast them
Liquidity preference	Investors risk-averse; long bonds carry more interest-rate risk and sacrifice liquidity → **premium**; normal curve is **positively sloped**
Preferred habitat (market segmentation)	Each investor has a preferred maturity; rates set by demand/supply in each segment; mismatch → higher rate to pull players out of habitat; curve can be **upward, falling or flat** (Saransh: explains almost any shape)

15 Convexity adjustment (SM 8.12)

Duration assumes a **linear** price–yield link → good only for small Δy. Adjustment = $C^* \times (\Delta y)^2 \times 100$, $C^* = (V_+ + V_- - 2V_0) \div [2V_0(\Delta y)^2]$

16 Convertibles, warrants, ZCB (SM 8.13–8.15)

- → **Convertible debenture** → equity after a period. Shares per debenture = **conversion ratio**; price paid per share = **conversion price**; **conversion value** = share price × shares per debenture
- → **Warrant** = right to subscribe equity during a period at a stated price; issued to **sweeten** a debenture issue. Value = **(MP − E) × n**
- → Warrant vs convertible: (i) warrant's conversion option is **detachable** → traded separately; (ii) warrant exercised for **cash**, convertible needs no cash
- → **Zero coupon bond** (TYK theory Q2): no coupon; issued at **discount**, one lump sum (face value = investment + accrued interest) at maturity; usually long maturity → long-range planning; issued by banks, government, companies — corporate ZCBs riskier (issuer strength, longer maturity) but higher return

MTP May 2025 extra terms (ICAI answer, not SM text): conversion parity = bond price ÷ ratio (5,300 ÷ 20 = 265); conversion premium; straight value (bond floor); downside risk; premium payback.

17 Refunding of bonds (SM 8.16)

- → Refund before maturity when rates **fall** — issue new lower-coupon bonds; hence **call feature**, which costs a **call premium** (repay above face value)
- → A strategic decision evaluated by capital budgeting — generally **NPV**: refund if PV of inflows (**net cash saving**) > PV of outflows (**call premium, interest during transition**, etc.)
- → New bonds usually issued **before** old ones are refunded → interest on **both** during the overlap

Exam method (app-written, not SM text): outlay = call premium (net of tax) + new issue cost + overlap interest (net) − tax saving on unamortised old issue cost written off; savings = interest saved (1 − t) ± issue-cost amortisation shield; discount at after-tax new k_d. SM TYK 25 has this set-up (₹9 m unamortised old cost, 4% call premium, 30% tax).

18 Money market instruments (SM 8.17)

Characteristics (5): short duration · large volume · de-regulated interest rates · highly liquid · safe (issuers' inherent financial strength).

Instrument	Key points (SM)
Call / notice money	Inter-bank; scheduled commercial banks; **call = overnight / 1 day**; **notice = 2–14 days**; manage day-to-day surpluses/deficits
Treasury bills	RBI; **91, 182, 364 days**; issued at discount; **Y = (F − P)/P × 365/M × 100**. TYK 28: 91-day @ 6% → **₹98.53**
Commercial bills	From genuine trade (credit sale); self-liquidating, negotiable; **3–6 months**; recourse to two parties
Certificate of deposit	Negotiable term deposits of banks from bulk depositors at market rates; demat or usance promissory note; issued at discount. E = (FV − SV)/SV × 12/M × 100 (SM eg ₹90 for 6 m → **22.22%**)
Commercial paper	Unsecured negotiable promissory note of high-rated companies for WC; **7 days–1 year**; at discount; India Jan 1990 (**Vaghul Committee**); bypasses banks; **no net increase** in borrowing (replaces WC limits). TYK 30: ₹10 cr, 91 days, 12.04% → net **₹9.70849 cr**
Repo / reverse repo	Sell securities + agree to repurchase ("ready forward"). Differences: (a) repo = RBI lends to banks; reverse = banks lend to RBI (b) repo from seller's (fund-raiser's) view (c) repo meets fund deficiency; reverse cuts excess liquidity (d) repo rate higher (e) repo contains inflation; reverse controls money supply

TYK 31 (repo): dirty = clean + accrued = 99.42 + 10 × 262/360 = **106.70**; 1st leg = 8 cr × 106.70/100 × 98/100 = **₹8,36,52,800**; 2nd leg × (1 + 5.65% × 14/360) = **₹8,38,36,604**.

19 Precautions before accepting (SM 10)

1. Good valuation is **not a precise estimate** — quality ∝ time spent collecting data and understanding the firm
2. More than financial statements — new businesses valued on **narrative** (scalability, replication, growth, cross-sell)
3. Find a **middle ground between numbers and story**
4. **Emotions** move stocks — ignoring them hurts consistency

4-MARK THEORY

20 Role of valuers (SM 9.1) — statutory purposes

1. **M&A / demergers / takeovers** — Income Tax Act, 1961 (tax payable)
2. **Slump / asset / IPR sale** — IBC, 2016 (liquidation; fair or liquidation value)
3. **Conversion of debt / security** — RBI (inbound / outbound foreign investment)
4. **Capital reduction** — SEBI ICDR / LODR / preferential allotment
5. **Strategic financial restructuring** — Companies Act 2013, SARFAESI 2002, Arbitration Act 1996; also Ind AS and court orders

21 Responsibilities (SM 9.2) — Model Code, Rule 12(e), Registered Valuers Rules 2017

1 Integrity & fairness — honest, true info, no disrepute · **2 Competence & due care** — standards, keep skills current, **can't disclaim liability** except for facts given by company · **3 Independence & disclosure** — no bias; no assignment if not independent; no dealing in subject's securities (PIT Regs 2015); no "mandate snatching" / "convenience valuations"; **no success fee**; declare past association · **4 Confidentiality**

5 Information management — contemporaneous records; co-operate in inspections; working papers **3 years** (or till tribunal case disposed) · **6 Gifts & hospitality** — none that affect independence · **7 Remuneration & costs** — transparent; only disclosed fees · **8 Occupation & restrictions** — not too many assignments; no employment unless certificate surrendered

22 Correlation — how Ch 5 actually connects

Ch 5 idea	The actual link (app-written cross-reference; Ch 5 side is SM)
$K_e = R_f + \beta(R_m - R_f)$	**Ch 6**: same CAPM; $\beta = Cov_{im} \div \sigma_m^2$ (variance, not σ). The K_e from Ch 6 is the discount rate in every DDM / FCFE here. Stock's ERP = $\beta \times$ MRP — not MRP.
$g = b \times r$	Gordon earnings model uses the same b and r: $P = EPS_1(1 - b) \div (K_e - br)$ is just $D_1 \div (K_e - g)$ with $D_1 = EPS_1(1 - b)$. If $r > K_e$, more retention raises value (Walter gives the same direction).
FCFF @ WACC, FCFE @ K_e	**Ch 13 / Ch 14**: firm value from FCFF − debt = equity value → ÷ shares = value per share → **swap ratio** = target's value per share ÷ acquirer's. A wrong discount rate (FCFE at WACC) flows straight into a wrong swap ratio.
Nominal CF ↔ nominal rate	**Ch 3 / capital budgeting**: (1 + nominal) = (1 + real)(1 + inflation). Never mix real cash flows with a nominal K_e.
Warrant (MP − E) × n	**Ch 9**: this is the **intrinsic value of a call** with strike E. A convertible = straight bond + embedded call on shares; bond floor = straight value.
Duration, immunization	**Ch 12**: interest-rate risk — price risk vs reinvestment risk are the same two risks; % price change ≈ −MD × Δy is how a 1% rate shock is sized. **Ch 8**: longer-duration debt / gilt funds swing more in NAV when rates move.
Repo, CP, CD, T-bill yields	**Ch 10**: money-market rates feed interest rate parity / money-market hedge (borrow or invest at these rates for the hedge period, days ÷ 360 or 365 as given).